

PHY199-RAP : Energy in the 21st Century

Course Description

This class will cover energy production and use in the 21st Century. We will address the physics behind energy generation of various methodologies, a historical overview of energy generation and consumption, the advent of anthropomorphic climate change, the full-stack development of supply chains and their impacts on various ethnic and social groups, geopolitical considerations of energy choices, the technological and sociological changes that impacted development in the 20th Century, and the new challenges of energy production in the dawn of the age of Artificial Intelligence while maintaining social justice for people everywhere.

Practical Information

Instructor: Dr. Salvatore Rappoccio (he/they). Call me “Sal”.

Lecture Hours: TBD

Lecture Room: TBD

Office Hours: TBD

Office: 335 Fronczak Hall

Telephone: 645-6250

e-mail: srrappoc@buffalo.edu

Course homepage

Our section is on <http://ublearns.buffalo.edu>.

Required Textbook

There is no required textbook. Course material will be based on handouts and lectures.

Accessibility Resources

If you are eligible for additional resources and require reasonable accommodations to enable you to participate in this course, such as note takers, readers, or extended time on exams, please contact the Accessibility Resources office, 25 Capen Hall (745-2608), and also see me during the first two weeks of class to give me your AR documentation.

Academic Integrity

Students are responsible for abiding by the academic integrity policies of the University. Academic dishonesty will be prosecuted to the fullest extent.

Use of Generative AI

The use of AI for your homework is not relevant as they are not graded. The quizzes and midterm are in-class, so you cannot use AI for them at all. You may use AI for study material as you wish. You can use AI for your final projects with citation of all **PRIMARY** source material, if you quantify the ecological and social implications of your use. This will be discussed in detail during class but will include the AI model you used, what company is providing the answer, the source material it is deriving its answer from, where the data centers for the AI are, and an estimate of the compute time to derive the answer). You will be asked probing questions during the Q+A sessions that you will be

expected to answer correctly, so if you do not understand the material you're presenting, you will not receive a passing grade.

Class Decorum

The main rule for class is consideration to all those within the class. No disruptive behavior will be tolerated. No texting or phone calls are allowed. Laptops, tablets, and cell phones may be used only for lecture notes and classroom participation software. Technology misused during class will not be allowed.

Communication Etiquette (e.g., email):

- Please use your university account for all correspondence. Emails from other accounts (e.g. gmail) will not be acknowledged.
- Use a clear subject line that includes the course name/section and the topic of the email such as "PHY199-RAP Question about the schedule". The easiest method is to use UBLearns directly.
- Please be respectful (i.e., not too casual) when addressing me in an email. Use an appropriate greeting (acceptable: "Dear Sal", "Dear Dr. Rappoccio", "Dear Professor"; unacceptable: "Hey there", "Yo Dawg"). Please proofread, spell check, and use complete sentences. This is so I can clearly understand what you are asking. Text-based communication is limited, so we all must do our best to assure proper information transfer. Keep the question(s) short and to the point. Show that you have made an effort to find the answer first in the text and lecture material (or even an outside source). State what you know in relation to what you are having a difficult time understanding.
- Allow 48 hours for a response to emails, excluding weekends and holidays.

Resignation Dates

Last day to drop a course without financial or academic penalty: Monday, 31-August-2026

Last day to resign a course: Wednesday, 11-November-2026

Available Resources on Sexual Assault

UB is committed to providing an environment free of all forms of discrimination and sexual harassment, including sexual assault, domestic and dating violence and stalking. You may call UB's Office of Equity, Diversity and Inclusion at (716) 645-2266 for more information or visit their website.

Counseling Services

As a student you may experience a range of issues that can cause barriers to learning or reduce your ability to participate in daily activities. These might include strained relationships, anxiety, high levels of stress, alcohol/drug problems, feeling down, health concerns, or unwanted sexual experiences. Counseling, Health Services, and Health Promotion are here to help with these or other concerns. You learn can more about these programs and services by contacting:

- Counseling Services:
 - o 120 Richmond Quad (North Campus), phone 716-645-2720
 - o 202 Michael Hall (South Campus), phone: 716-829-5800
- Health Services: Michael Hall (South Campus), phone: 716- 829-3316
- Health Promotion: 114 Student Union (North Campus), phone: 716- 645-2837

UB Portfolio

If you are completing this course as part of your UB Curriculum requirements, please select an ‘artifact’ from this course that is representative of your learning and save it in a safe location with a clear title. Your final UB Curriculum requirement, UBC 399: UB Curriculum Capstone, will require you to submit these ‘artifacts’ as you process and reflect on your achievement and growth through the UB Curriculum. Artifacts include homework assignments, exams, research papers, projects, lab reports, presentations, and other coursework. For more information, see the UB Curriculum Capstone website: <https://www.buffalo.edu/ubcurriculum/capstone.html>.

Grading

Grades on this course will be based on your scores on participation in classes, one midterm exam, in-class quizzes, and a final project as follows:

- Participation: 20%
- In-class quizzes: 20%
- Midterm score: 30 %
- Final presentation: 30 %

Letter grades will be assigned based on your cumulative numerical grade, curved to the activity of your class. You will never be “curved down”, so if everyone gets an A, then that’s AHH-mazing! (See what I did there? It’s a pun. Folks... it’s a pun. Get it? Never mind.)

Final Grades:

Grade	Quality Points	Percentage		Grade	Quality Points	Percentage
A	4.0	93.0 -100.00		C	2.00	73.0 - 76.9
A-	3.67	90.0 - 92.9		C-	1.67	70.0 - 72.9
B+	3.33	87.0 - 89.9		D+	1.33	67.0 - 69.9
B	3.00	83.0 - 86.9		D	1.00	60.0 - 66.9
B-	2.67	80.0 - 82.9		F	0	59.9 or below
C+	2.33	77.0 - 79.9				

Absence Policy

You are adults. I trust you. The grading system is built to account for all absences such as illness, changed work shifts, death in the family, surprise grandparent visits, dog eats your homework for a different class, the occasional earthquake or other natural disaster, or “just can’t even so I need a mental health day”. If you require more than those number of absences for an extended illness or emergency, please contact me as soon as possible to make special arrangements.

Participation (20%)

You will be graded on participation, which will be quantified as follows. Questions will be asked during class, and you are expected to answer via a classroom management system (tbd). You are only graded on participation and are encouraged to discuss with your peers. You will receive a 100% score if you answer 80% of the questions in the class to account for absences (see the Absence Policy). If you answer more than 80% of the questions, you are still graded for the highest 80%, raising your overall average.

Homework and In-Class Quizzes (20%)

Homework questions will be assigned but not graded. Instead, in-class quizzes will be administered frequently based on the homework questions. Both may consist of multiple choice or short answer questions.

In-Class Quizzes will be regularly offered in class. These are based on the homework questions. You will only be graded for the highest 80% of your quizzes and the lowest 20% of your quizzes will be dropped to account for absences. If you attend more than 80% of the classes, you still are graded for the highest 80%, raising your overall average.

Midterm (30%)

There will be a Midterm Exam that will cover questions similar to your In-Class Quizzes and Homework. They may be multiple choice or short answer. If you miss the midterm, you will need to take a make-up exam.

Final presentation (30%)

Each student will be expected to prepare a 10 minute presentation as a final exam with a 5-10 minute Q+A session. You should pick a topic that is relevant to the class material. You are welcome to include a 1-3 minute social media video that you create (without AI) that will be part of your allotted 10 minutes. You will be graded as follows:

- Proposal (5%)
- Slides of the talk (10%)
- Delivery of the talk (10%)
- Answering questions (5%)

The proposal will be due one month before the presentations begin. You will be provided feedback on the topic and expected to address such questions in your talk. Everyone will upload their slides ahead of the presentations so you all have equal time to prepare. I will randomly pick the order of the presentations and tell you ahead of time. You are required to attend as many of your peers' presentations as possible and this will be counted as part of your Participation grade (see Absence Policy above).

Course Schedule

This is the course schedule. There is intentionally some extra space to discuss special topics in the last few weeks, designed to help you with your presentations in case you would like to learn more about specific topics. We will vote on these topics in class. In-class time will be allocated in the week of Thanksgiving break for you to receive direct feedback from your peers and myself. The last 2-3 weeks of class will be your presentations. You will not have a group, each presentation will be your own work.

Week number	Topic	Material	Assignment
1 (24-Aug)	Introduction	Syllabus, Lecture notes	Syllabus quiz
2 (31-Aug)	Energy generation	Lecture notes on electricity and magnetism , nuclear fission and fusion , Levelized Cost , the grid . Skills taught: Note-taking.	Academic Integrity at UB course on UBLearns , with quiz. Share badge on UBLearns
3 (7-Sep)	Political implications of energy generation	Lecture notes on the AC/DC Current Wars , anthropogenic climate change , Battle of Blair Mountain	Quiz on note-taking and material from weeks 1+2
4 (14-Sep)	Political implications of energy generation	Lecture notes on the death cost of electricity generation , public health costs of carbon emissions . Skills taught : time management.	Quiz on material from week 3
5 (21-Sep)	Human costs of energy generation	Lecture notes with excerpts from Hiroshima , The History of Uranium Mining and the Navajo People , The Love Canal Story Is Not Finished .	Quiz about films and time management skills. Options: * Hiroshima : The Aftermath * Fat Man and Little Boy * Oppenheimer

Week number	Topic	Material	Assignment
6 (28-Sep)	Human costs of energy generation	Lecture notes on impacts of mining rare earth materials for batteries , disposal of solar panels . Skills taught: Research Skills.	Quiz about material from weeks 5+6
7 (5-Oct)	Computing and AI	Capability of generative AI, water cost , chip production , the AI bubble , automation in the workplace	Quiz on Research skills
8 (12-Oct)	Future of energy generation	Population predictions, requirements for needs	Research proposal due
9 (19-Oct)	Review	Review	Midterm
10 (26-Oct)	How to give a good talk	Lecture notes on presentations to different audiences	
11 (2-Nov)	How to communicate to non-specialists	Lecture notes on scientific communication to non-specialists.	Bibliography for presentation due
12 (9-Nov)	Special topics : voted by class	Lecture notes on topics relevant to presentations of the students.	Quiz on material from weeks 10+11
13 (16-Nov)	Special topics : voted by class	Lecture notes on topics relevant to presentations of the students.	First draft of presentation due
14 (23-Nov)	In-class work on Presentations	In-class work on Presentations	Thanksgiving break
15 (30-Nov)	Presentations	Presentations	Presentations
16 (7-Dec)	Presentations	Presentations	Presentations
17 (14-Dec)	Presentations	Presentations	Presentations

Learning Outcomes

The UB Seminar teaches new students foundational skills for college success through disciplinary knowledge and tools. It explores the academic expectations pertaining to being a student at UB and to higher learning at a research university. The seminar emphasizes the critical thinking, ethical reasoning, and reflective discussion essential for civic discourse.

SLI Program Student Learning Outcome	Examples of how this specific LO is met in this course	Instructional delivery	Assessment technique
Demonstrate critical thinking using multiple modes of inquiry and expression.	Students will investigate historical debates related to energy generation and learn how they still apply to the challenges of the 21st century.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project
Understand and apply methods of analysis, synthesis, and critical reading.	Students will investigate energy generation at the mechanical, atomic, and nuclear levels, and how that energy is transferred across the electric grid to provide power. This will be used to correctly assess pros and cons of energy generation and the challenges to be faced in the future.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project
Cultivate effective research and study habits, including note-taking and time management, through engagement with course content.	Students will have low-stakes in-class quizzes as well as participation-only discussions among their peers to show how to effectively manage time and course load.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project

SLI Program Student Learning Outcome	Examples of how this specific LO is met in this course	Instructional delivery	Assessment technique
Identify and analyze important disciplinary debates and engage in deliberative discourse.	We will discuss a number of large-scale scientific and engineering developments and their impacts on power generation, including the AC/DC “wars”, the Manhattan Project, Artificial Intelligence, material science, and more.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project
Recognize and engage with ethical issues pertaining to the subject matter.	Investigations of ethical and geopolitical consequences of mining, water usage, power generation, waste management, the “death cost”, and other areas.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project
Demonstrate an understanding of academic integrity.	Students will be allowed to use external resources (including AI) in their coursework, with proper citation and estimation of the real cost of using generative AI for their coursework.	Lecture, assigned readings, supplementary videos	Quizzes, midterm, project